Astonishingly, the latter has the same accuracy as the other 2, even if it misses every single point in the false category.

> **TIP:** if you want to prove it by yourself, plug in these numbers in the formula for accuracy

In machine learning, if you were to use this metric on a training set skewed in a similar way[3]... then you'd get a terrible model, or more precisely a model that is likely to generalize poorly.

F.4.2 Precision and Recall

To improve over bare accuracy, we need to keep track of the info about each category separately.

To this extent, we can define two new metrics:

- *precision* (also called positive *predictive value*) defined as the ratio of correctly predicted true points (the *true positives*) over the total number of points for which true was predicted:

$$precision = \frac{T_P}{P_P}$$

- *recall* (also known as *sensitivity*) defined as the fraction of *true positives* over the total number of actual positives:

$$recall = \frac{T_P}{N_P}$$